

Nicholas Lee

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Education

University of Waterloo – BAsC in Mechatronics Engineering, GPA: 3.85

Expected May 2028

Skills

Design & Manufacturing: SolidWorks (CSWA), Onshape, GD&T, sheet metal, DFM/DFA, 3D printing, milling, laser cutting, Kenesto PDM, nTop

Simulation & Analysis: COMSOL, ANSYS, CFD, FEA

Programming: Python, MATLAB/Simulink, Ladder Logic (Siemens TIA Portal), Git

Work Experience

Research Intern, Harvard University, Aizenberg Lab – Cambridge, MA

May 2026 – Aug 2026

- Engineered watertight nasal airway models of real animal anatomy from CT-scanned skulls for future benchtop experiments, segmenting nasal cavities in ITK-SNAP and repairing meshes in MeshLab and Meshmixer
- Designed a parametric COMSOL model of a nasal-inspired channel driven by transient sniff waveforms, and ran six studies sweeping 19 geometries, 13 sniff conditions, and 10 chemical species for an adaptive electronic nose
- Defined optimal sensor placement and sniff operating ranges for the e-nose by cross-validating detection results across independent parameter sweeps
- Streamlined post-processing for six transient COMSOL studies (700+ cases, 200K+ rows) with a Python pipeline (Pandas, NumPy, Matplotlib) calculating 11 metrics, 80+ figures, and 16 tables in one script

Undergraduate Research Assistant, MSAM Lab – Waterloo, ON

Jan 2026 – Apr 2026

- Delivered motor-driven two-stage gearbox demonstrators used as live hardware in undergraduate lectures by iterating FDM-printed gears, shafts, and housings until assemblies ran smoothly
- Accelerated helical gear design by building a parametric SolidWorks model for rapid geometry changes
- Reduced gearbox-housing material use while preserving structural strength by designing internal lattice infill structures in nTop for additive manufacturing

Engineering Intern, Refresco Beverages – Mississauga, ON

Jan 2025 – Aug 2025

- Redesigned a pneumatic can-rejection piston mount and cover in SolidWorks to integrate a replacement vendor piston and shield the rod from the syrup splash behind monthly failures, delivering a full GD&T fabrication package
- Initiated a \$100K CAD CAPEX project to reduce worker injuries by scoping vacuum lift-assist solutions, evaluating vendor proposals, and presenting the business case to the plant manager
- Improved plant uptime by implementing preventive maintenance procedures and updating drawings and BOMs in the CMMS

Student Design Teams

Mechanical Team Member, Midnight Sun Solar Car Team – Waterloo, ON

Sep 2025 – Present

- Reduced ballast box mass by 20% by designing an aluminum sheet-metal assembly in SolidWorks, verifying strength under turning and braking loads through static FEA; the box is now on the competition vehicle
- Designed two assembly fixtures: surface-modeled wheel-cover stencils referenced to vehicle datums, and a tool-free press-fit camera mount with fit testing over multiple print iterations

Mechanical Designer, Waterloo Automation Collective – Waterloo, ON

Jan 2026 – Present

- Led the clamping subsystem for an injection molder from concept through detailed CAD and prototyping; parts now being milled for first assembly
- Enabled rapid geometry iteration and centralized configuration management by developing parametrically driven clamping linkages in Onshape and SolidWorks